

How Useful Is a Learned World Model?

Planning, Replication, and Model-Quality Diagnostics in a Multi-Agent Evolutionary Sandbox

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ABSTRACT

We present a self-contained **multi-agent evolutionary sandbox** in which populations of recurrent neural agents forage, plant, and reproduce in a procedurally generated 2-D ecology with terrain, rivers, seasons, weather, and soil dynamics. Agents learn *within their lifetime* by PPO with full-network backprop through an attention encoder and a GRU, and pass their trained weights to offspring (Lamarckian inheritance), so natural selection acts directly on learned behaviour. Three brain generations share one append-only genome contract and can compete in a single world, and the live state streams to three visualizations (2-D, isometric, and an in-browser voxel client). The sandbox functions as an ecology-as-benchmark: capabilities are *observed* in a living population rather than rewarded into existence. We report measured results at every level. **Ecology:** populations self-regulate to a carrying capacity independent of founder count (8/24/48 founders converge within ~ 200 ticks). **Architecture selection:** in shared-world competition an attention brain displaces a legacy recurrent brain, growing from 1/8 founders to 96% of the population (mean fitness 10.57 vs 8.95). **World models:** an offline observation-space model trained on 291k logged transitions beats a static-world baseline by 20% (held-out Δ o MSE 0.0268 vs 0.0337) and explains $\approx 67\%$ of reward variance; a per-agent latent dynamics head — though only marginally better than chance — flips the population from aimless wandering to a goal-directed forage $\rightarrow$ eat $\rightarrow$ plant strategy when used for cheap planning and curiosity (exploration $\times 2$, peak fitness $+49\%$). We then use the sandbox for what such a testbed is for: a **replication study** of progressively heavier model-based planning (policy-guided rollouts, cross-entropy-method search, Dreamer-style imagination, warmup scheduling, and error-gated scheduling), instrumented with a k-step open-loop rollout-error diagnostic, a measured-readiness planner gate, and a multi-step consistency training loss. Under 4-seed replication only the cheapest upgrade survives (policy-guided rollouts: final fitness 51.8 ± 3.7 vs 45.3 ± 13.7 at equal speed); the heavier machinery does not earn its cost under any schedule, fixed or measured — and the instrumentation lets us attribute that outcome to compounding model error rather than to the schedule (open-loop error is $2.3\times$ worse early and falls 56% over 6,000 ticks of training; the multi-step loss reduces it exactly where the model is worst, 6.09 ± 0.11 vs 6.56 ± 0.09 at tick 1,000). The broader lessons: a weak model can be genuinely useful for exploration and action ranking; sharper search on the same weak model is worse than blunt search; single-seed effect sizes in ecological simulations routinely evaporate under replication; and the model's open-loop error is the quantity to measure before trusting any of it. All experiments are reproducible end-to-end, and every per-run number appears inline in this paper.

Keywords: multi-agent reinforcement learning · world models · model-based planning · replication · model error · intrinsic motivation · open-ended evolution

1. Introduction

Reinforcement-learning agents are usually trained one policy at a time against a fixed task, with a learned "world model" either absent or assumed helpful. This work instead studies *populations* of agents that live, learn during their lifetime, and reproduce with inheritance of learned weights (a Lamarckian scheme), so selection and individual learning act together and behaviour is *observed* rather than rewarded into existence. Into this sandbox we introduce a per-agent latent world model and ask, with matched A/B runs and multi-seed replication, what it is actually good for. The contributions:

1. **A complete, reproducible sandbox** — procedural ecology, three brain generations under one append-only genome

contract, PPO with full-network backprop through a GRU, Lamarckian inheritance, and three live visualizations (2-D, isometric, browser voxel).

2. **Measured capabilities:** architecture selection by ecological competition (Sec. 6.1); an offline world model with quantified accuracy (Sec. 6.2); a planning+curiosity phase change in behaviour (Sec. 6.3).
3. **A replication study of planner upgrades** (policy-guided rollouts, CEM, imagination, warmup scheduling) whose headline is honest: most single- and two-seed wins did not survive four seeds; policy-guided rollouts did (Sec. 6.4–6.6).
4. **A model-quality toolkit** motivated by the failures: a k-step open-loop rollout-error diagnostic computed during training, an error-threshold readiness gate for model-based planners,

and a multi-step consistency training loss — with measurements (Sec. 6.7).

2. Related work

World models and latent imagination. Ha and Schmidhuber [1] train agents inside a learned generative model; PlaNet [3] and the Dreamer line [2,4] learn latent dynamics and optimise behaviour on imagined rollouts. Our offline population model is the [1]-style predictor; the per-agent latent head is the Dreamer-style component the planner, curiosity, and imagination losses consume.

Decision-time planning. Random shooting and PETS [5] act on the best of many sampled model rollouts; MPPI [6] reweights trajectories; MuZero [7] runs policy/value-guided search; MBPO [8] uses short model rollouts as training data. Our controllers are the simplest instances of shooting, policy-guided shooting, and categorical CEM.

Policy learning and curiosity. Lifetime learning is PPO [9] with GAE(λ) [10] (a legacy A2C/A3C [11] learner exists); intrinsic reward is forward-model prediction error, the ICM signal [12] (related: RND [13]). *Perception, memory, evolution:*

attention [14], GRU [15], fission-with- mutation neuroevolution [16], novelty metrics [17].

Model error. The failure mode we measure — a planner exploiting the compounding open-loop error of a model trained one step at a time — is the objective-mismatch problem [19]; our multi-step consistency loss follows the data-as-demonstrator insight of training on the model's own predictions [18].

3. The sandbox

3.1 World generation

Each world is procedurally generated (Fig. 1): layered value noise $\rightarrow$ an elevation field; downhill tracing from high-elevation sources carves rivers; biomes (soil/sand/water/rock) derive from elevation plus a moisture field with fertile corridors along rivers; an ecology pass seeds plants that grow through seed $\rightarrow$ sprout $\rightarrow$ mature stages, fruit into berries, and drop seeds that germinate under a neighbourhood carrying-capacity cap (≤ 3 plants within radius 2). The default experimental arena is 64×64 with soil/rock/water/sand ratios 0.62/0.18/0.12/0.08.

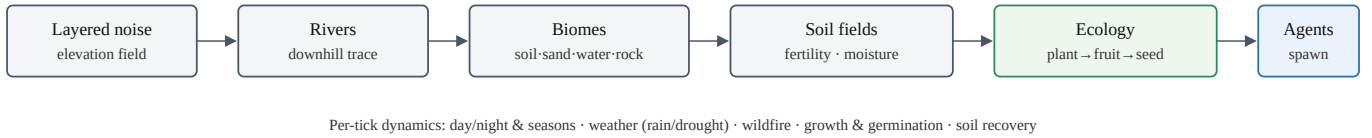


FIGURE 1. Procedural world-generation pipeline and the per-tick ecological dynamics layered on top.

3.2 World dynamics

Per tick: a day/night cycle (length 200) modulates light; seasons (length 2,000) modulate temperature; weather switches between rain (recovering soil moisture) and drought (evaporation $\times 2$); optional wildfire ignites, spreads against a moisture threshold, and returns nutrients; soil fertility/moisture relax slowly toward equilibrium; plants consume fertility and moisture as they grow. Agents pay a metabolism cost scaled by temperature, eat berries (+20 energy), can plant seeds (−5 energy), and die at zero energy or `max_age` 1,000, returning 15% of consumed fertility to the soil. Water is impassable. Reproduction is energetic: above 45% of max energy and age ≥ 50 , an agent splits 40% of its energy into an offspring whose genome is the parent's

trained weights plus Gaussian mutation ($\sigma=0.02$), capped at 30 alive agents in the experimental arena.

3.3 Observability: three live views

The same live world is observable three ways (Fig. 3), all fed by one read-only state bridge that never affects dynamics: a top-down 2-D biome/elevation map, an isometric 2.5-D render, and an interactive in-browser voxel client streamed over server-sent events (terrain columns, trees that scale with plant maturity, berries, seeds, agents with follow-markers and vision-grid overlays, weather, and a live HUD reporting the population's brain mix). Every study below was spot-checked visually in at least one viewer.

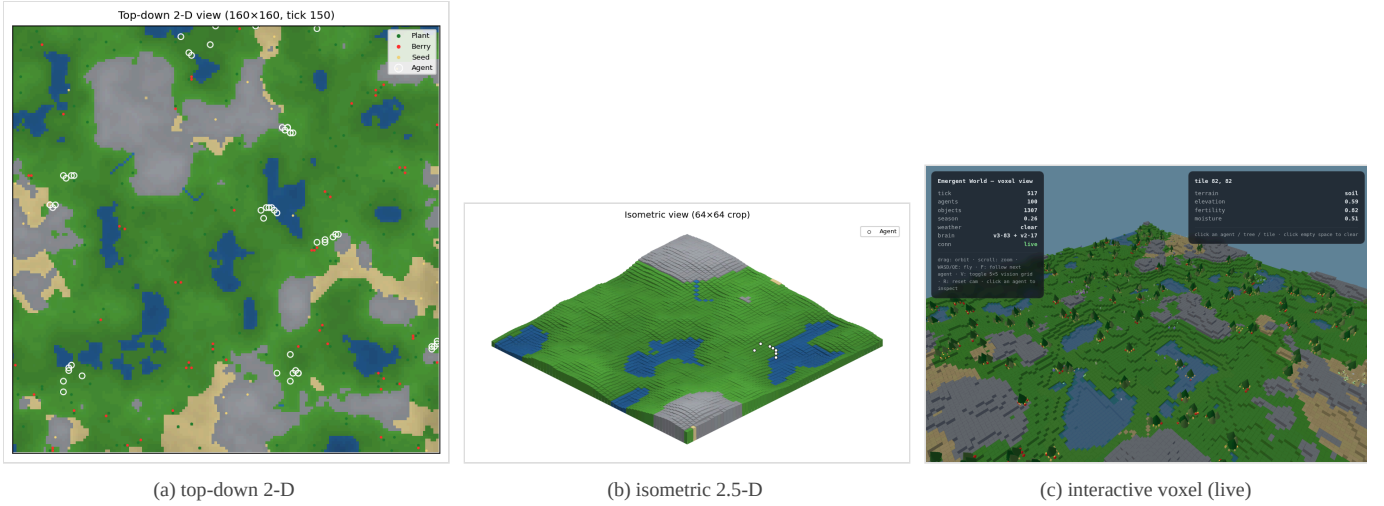


FIGURE 3. Three views of the *same* simulation, all from the read-only state bridge: (a) data-driven top-down map (160×160), (b) isometric render of a 64×64 crop, (c) the live browser voxel client.

4. Agents

Agents act once per tick on an egocentric, facing-rotated 5×5 vision grid plus internal state, stimulus features, inventory, and

(v3.5) social/climate features. Three architectures share one genome/IO contract (Table 1); the genome layout is *append-only*, so a saved population trained under one version migrates forward and can compete in a shared world.

TABLE 1. The three agent architectures (shared append-only genome/IO contract).

Brain	Obs	Actions	Params	Core
v2 (legacy)	72	8	≈8.9k	dense encoder → GRU → policy/value
v3 (attention)	72	8	≈17.3k	tokenised vision + attention → GRU → [z,h] value
v3.5 (social)	78	9	≈21.1k	v3 + 6 social/climate inputs + SIGNAL + pheromones

4.1 The v3 / v3.5 brain

Figure 2 shows the data flow. Each of the 25 vision tiles is embedded by one shared matrix (position codes appended — position-equivariant perception at constant parameter count); a

single attention query derived from the agent's internal state pools the tiles [14]; the latent $z=[state \mid pooled\text{-}vision]$ feeds a GRU [15] memory h ; the policy head reads h , the value head the joint $[z,h]$.

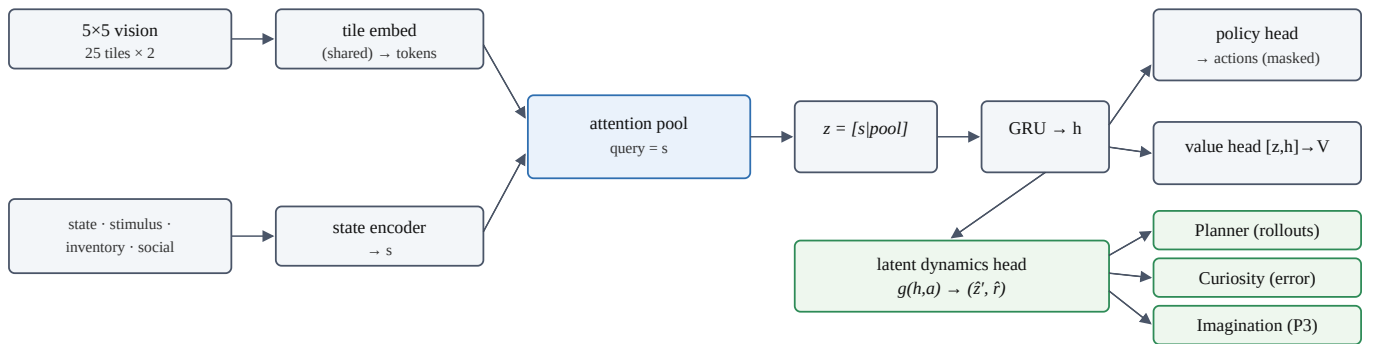


FIGURE 2. The v3/v3.5 agent brain. Green blocks are the optional latent world model and its three consumers — planner (Sec. 5), curiosity, and the imagination actor-critic (Sec. 5.4).

4.2 Lifetime learning and inheritance

PPO [9] on stored sequences (seq_len 8, GAE $\lambda=0.95$, clip 0.2, entropy 0.01, lr 3×10^{-4} , 1 epoch per update) with **full-network backprop**: gradients flow through the policy and value heads,

the GRU through time, and the attention encoder. A global scheduler staggers updates across ticks. Trained weights are written back into the genome (**Lamarckian**), so offspring inherit learned behaviour [16]. Newborns get fading instinct

biases (eat-when-hungry, approach-food) that decay to zero by age 150.

4.3 The per-agent latent world model

When enabled, the genome grows a dynamics head $g(h,a) \rightarrow (\hat{z}', \hat{r})$ — a one-hidden-layer MLP (width 32) from the GRU hidden state and a one-hot action to a predicted next latent and reward, trained as an auxiliary loss in the same PPO update (targets stop-gradient). It powers *planning* (Sec. 5), *curiosity* (ICM-style z-scored prediction error [12]), and *imagination* (Sec. 5.4).

4.4 The offline population world model

Separately, a population-level model $f(o,a) \rightarrow (\Delta o, \hat{r}, p_{done})$ (two tanh layers, 128 hidden) trains offline by regression on logged transitions from full runs [1], giving a measurable reference for what these small models can learn (Sec. 6.2).

5. Planning methods

The planner is receding-horizon model-predictive control in latent space: from the current hidden state h , imagine S candidate action sequences of depth D through the dynamics head, score each by $\sum_k \gamma^k \hat{r}_k + \gamma^D V$, execute the best first action, and (by default) replan next tick. All variants are config-gated and off by default; the defaults reproduce the legacy controller exactly. Experiments use depth 2, 6 samples.

5.1 Shooting (baseline). First actions uniform over the valid-action mask; continuations uniform-random [5].

5.2 P1 — policy-guided rollouts. Continuations sampled from the agent's *own policy* at the imagined latent state (Dreamer-style [2]): in-distribution, lower-variance rollouts. Optional first-action biasing, reward/value z-scoring, and plan commitment are implemented; P1 experiments use uniform first actions, no normalisation, commit 1.

5.3 P2 — cross-entropy method. A categorical distribution per step, refined over 3 iterations: sample the population, keep the top-30% elites, refit smoothed action frequencies [5,6]. TD(λ) scoring implemented ($\lambda=1$ used below).

5.4 P3 — imagination actor-critic. Dreamer-style [2]: from detached real hidden states the actor rolls 5 steps in the dynamics head; TD(λ) returns from the critic score the

trajectory; REINFORCE with a value baseline trains the actor, regression the critic. Planning is *distilled into the policy* instead of paid per tick.

5.5 Warmup scheduling. At tick 0 the model is untrained, so model-heavy planning "shoots off" a garbage model. The planner can run a cheap exploratory strategy (P1) until `warmup_ticks`, then switch to the configured strategy; imagination is gated the same way.

5.6 Model-quality toolkit (this work). Three mechanisms motivated by Sec. 6.5–6.6: **(i) a k-step rollout-error diagnostic** — every learner update, roll the dynamics head *open-loop* for k steps from real hidden states (its own predictions fed back through the GRU, real logged actions) and measure latent MSE against the real encoded latents at each horizon: exactly the compounding-error regime the planner uses, measured on-line per agent at negligible cost and logged per generation; **(ii) readiness gating** — the planner switches to its main strategy when the measured error EMA drops below a threshold (latched; `warmup_ticks` becomes a switch-anyway deadline) instead of on a blind tick count; **(iii) a multi-step consistency loss** — the standard auxiliary loss trains one-step prediction only, leaving compounding error unconstrained [18,19]; this loss rolls the head open-loop for k steps on stored sequences and penalises latent/reward error at horizons $2..k$.

6. Experiments and results

Throughout: *peak fitness* is the highest per-generation mean agent fitness in a run; *final* is the last generation's mean; *tiles* is unique tiles visited per agent; *EAT* counts successful eats; *seeds* counts seeds planted; ticks/s is single-core throughput including learning. All planning runs: 64×64 world, v3.5+PPO, world-model head on, curiosity off, cap 30, generation length 1,000. Fitness is the simulation's internal proxy; comparisons are within-simulation.

6.1 Ecological competition between architectures

One 160×160 world, 8 founders (1 v2, 7 v3), offspring breed true, 4,000 ticks, cap 100. The attention brain displaces the legacy brain — final population {v3:96, v2:4} (Table 2). Both cohorts reach 100% eating success, so the edge is foraging efficiency and reproduction, not basic feeding.

TABLE 2. Cohort comparison from a single shared 4,000-tick run (lifetime totals).

Cohort	Agents	Mean age	Max age	Mean fitness	Eat succ.
v2 (legacy)	255	98.3	259	8.95	100%
v3 (attention)	1715	101.3	493	10.57	100%

6.2 Accuracy of the offline world model

291,300 transitions logged from a v3.5+PPO run (64×64, pop 30; all nine actions exercised, SIGNAL 67,027 times). Training:

20 epochs, loss 3.20 → 1.71 (Fig. 4). Held-out accuracy in Table 3. The model is real but modest — **20% better than**

"**nothing changes**" — a number to keep in mind for everything that follows.

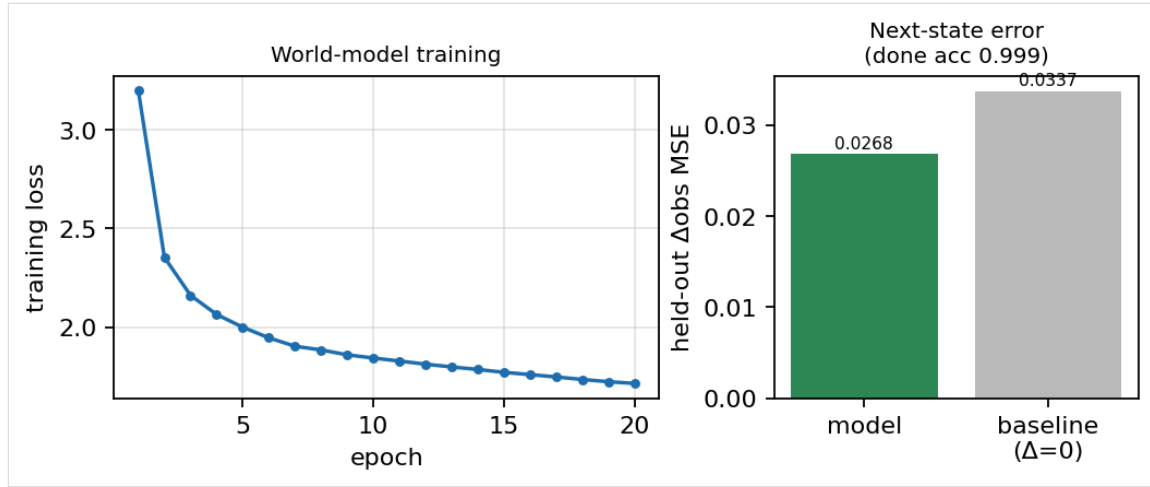


FIGURE 4. World-model training loss (left) and held-out next-state error vs. the predict- $\Delta=0$ baseline (right).

TABLE 3. Held-out predictive accuracy of the population world model.

Metric	Model	Baseline / scale	Reading
Δo MSE	0.0268	0.0337 ($\Delta=0$)	20% lower error
Reward MSE	1.70	variance 5.15	$\approx 67\%$ variance explained
done acc.	0.999	—	termination solved

6.3 Planning + curiosity: a behavioural phase change (single seed)

Matched A/B (seed 42, 3,000 ticks): identical worlds and learning; the treatment adds the dynamics head + shooting planner + curiosity (Table 4). Behaviour flips from aimless turning to the forage $\rightarrow$ eat $\rightarrow$ plant loop; per-agent strategy

entropy *rises* while pairwise novelty *falls* ($0.320 \rightarrow 0.092$): agents independently converge on the same effective strategy [17]. This is the honest headline for a weak model: it doubled exploration and halved idling *despite* being only 20% better than chance, because ranking actions and flagging surprise demand far less of a model than accurate simulation does.

TABLE 4. Matched A/B (single seed 42): planning + curiosity vs. baseline, v3.5+PPO, 3,000 ticks.

Metric	Baseline	+Plan&Curiosity	Δ
Avg peak fitness	38.7	57.8	+49%
Mean fitness (final)	40.8	70.7	+73%
Avg lifespan (ticks)	422	548	+30%
Mean energy (final)	109	165	+51%
Tiles explored / agent	27.7	54.9	+98%
EAT attempts	185	1196	$\times 6.5$
Seeds planted	226	929	$\times 4.1$
Turning (L+R) share	51%	36%	-15pt
Strategy entropy	1.61	2.44	+0.83
Pairwise novelty	0.320	0.092	converges

6.4 The planner ladder, single seed: an apparent staircase

Matched single-seed A/Bs (seed 42, 2,000 ticks; Table 5): each rung of model-based sophistication appears to buy more fitness

— P1 +21%, P2 +32% over the baseline; and in a separate A/B with the planner off in both arms, the P3 imagination loss appears to add +25% peak fitness ($32.6 \rightarrow 40.6$), +38% planting,

and -16pt idle waiting for a 20% training-time cost. Sec. 6.5 is why face value is not enough.

TABLE 5. Single-seed planner ladder (seed 42, 2,000 ticks, depth 2, 6 samples) and the P3 imagination A/B (planner off in both arms).

Variant	rollouts/decision	peak	EAT	seeds	WAIT	ticks/s
shooting (baseline)	6	33.8	306	230	21.9%	6.54
policy_shooting (P1)	6	41.1	341	289	22.6%	6.35
cem (P2)	18 (3×6)	44.5	487	396	26.9%	5.19
imag off	0	32.6	458	61	31.0%	9.05
imag on (P3)	0	40.6	187	84	15.4%	7.22

6.5 Four-seed replication: the staircase collapses

The identical five arms re-run at seeds 1–4 (2,000 ticks; 20 runs). Table 6 gives every run; Table 7 the aggregates; Fig. 5 the picture. Three findings: **(i) P1 survives replication** — policy-guided rollouts beat the baseline on mean peak (43.5 vs 41.1), on final fitness with far lower variance (51.8 ± 3.7 vs 45.3 ± 13.7), on eating and planting, at equal speed; the

mechanism is cheap and plausible (in-distribution continuations cut rollout variance without extra rollouts). **(ii) P2's +32% was one seed** — across seeds CEM ties the baseline on peak (41.1 ± 6.4), is clearly worse on final fitness (34.5 ± 9.9), and runs ~23% slower; its seed-1 win (51.9) is bracketed by losses at seeds 2–4. **(iii) P3's +25% was noise** — imagination on vs off: 35.3 ± 5.2 vs 34.8 ± 2.8 , at a 22% throughput cost.

TABLE 6. Four-seed replication, complete per-run data (2,000 ticks per run).

seed	variant	peak	lifespan	tiles	EAT	seeds	final	ticks/s
1	shooting	33.15	409.0	40.7	108	156	30.05	5.57
1	policy_shooting	43.12	453.2	50.4	562	318	55.96	6.90
1	cem	51.89	503.3	49.7	640	395	51.02	4.96
1	imag_off	33.74	395.4	14.6	65	326	41.51	8.37
1	imag_on	28.10	384.9	28.9	123	229	27.11	6.58
2	shooting	41.25	463.9	38.9	328	283	35.26	6.62
2	policy_shooting	45.26	449.2	38.6	504	519	47.69	6.15
2	cem	39.67	456.7	47.8	263	428	26.75	5.10
2	imag_off	31.52	364.5	16.5	125	101	22.04	8.85
2	imag_on	32.65	436.4	13.1	156	63	36.36	6.62
3	shooting	45.74	467.6	39.8	738	383	51.23	6.76
3	policy_shooting	48.34	478.9	40.2	779	393	48.40	6.69
3	cem	35.10	446.3	33.3	174	179	27.10	5.29
3	imag_off	34.49	430.0	41.5	328	274	45.25	8.81
3	imag_on	41.22	443.0	19.5	153	252	36.54	6.85
4	shooting	44.42	466.3	46.0	355	320	64.81	6.92
4	policy_shooting	37.32	438.8	42.2	248	185	54.98	5.99
4	cem	37.66	426.9	49.5	254	268	33.25	4.73
4	imag_off	39.32	426.9	31.2	242	147	46.03	8.33
4	imag_on	39.10	397.5	51.4	440	551	39.32	6.85

TABLE 7. Replication aggregates (mean $\pm$ std over 4 seeds).

Variant	peak fitness	final fitness	EAT	seeds	ticks/s
shooting	41.1 $\pm$ 4.9	45.3 $\pm$ 13.7	382 $\pm$ 227	286 $\pm$ 83	6.5
policy_shooting (P1)	43.5 $\pm$ 4.0	51.8 $\pm$ 3.7	523 $\pm$ 189	354 $\pm$ 121	6.4
cem (P2)	41.1 $\pm$ 6.4	34.5 $\pm$ 9.9	333 $\pm$ 181	318 $\pm$ 100	5.0
imag_off	34.8 $\pm$ 2.8	38.7 $\pm$ 9.8	190 $\pm$ 102	212 $\pm$ 91	8.6
imag_on (P3)	35.3 $\pm$ 5.2	34.8 $\pm$ 4.6	218 $\pm$ 129	274 $\pm$ 176	6.7

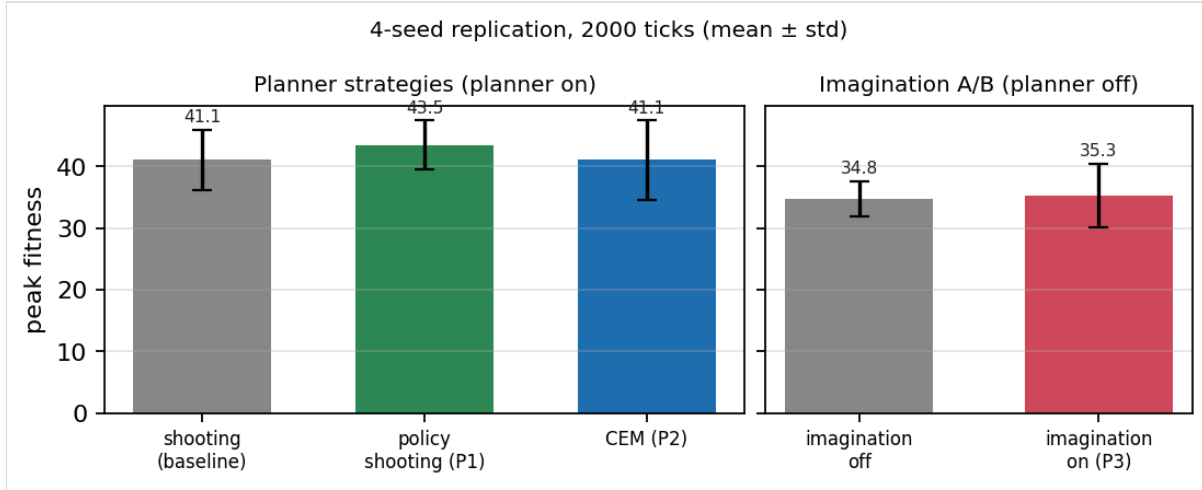


FIGURE 5. Four-seed replication of the planner ladder (peak fitness, mean $\pm$ std). Left: planner strategies. Right: the imagination A/B (planner off in both arms). Only P1 separates from its baseline.

6.6 Warmup scheduling: hypothesis, pilot win, honest reversal

Hypothesis. P2/P3 fail from a *cold start*: at tick 0 the dynamics head is random, so CEM optimises fantasies and imagination trains the actor on garbage. Gate them on model training time

and they should work. **Pilot (2 seeds, 6,000 ticks; Table 8):** *scheduled* (P1 until tick 5,000, then CEM+imagination) beat the baseline on both seeds (77.9 > 74.5; 55.1 > 41.1), mean peak 66.5 $\pm$ 11.4 vs 57.8 $\pm$ 16.7, while cold-start CEM+imagination was a coin flip. The hypothesis looked confirmed.

TABLE 8. Warmup pilot, complete per-run data (2 seeds, 6,000 ticks). *naive* = CEM+imagination from tick 0; *scheduled* = P1 until 5,000 then CEM+imagination.

seed	variant	peak	lifespan	tiles	EAT	seeds	final	ticks/s
1	baseline	74.47	721.8	64.0	2225	1818	51.69	5.10
1	naive	52.35	620.5	70.5	925	1015	40.77	3.24
1	scheduled	77.90	701.6	85.1	2372	2532	46.19	4.35
2	baseline	41.11	415.2	42.9	1805	1394	41.30	5.23
2	naive	73.34	713.1	64.4	2047	1260	52.92	3.36
2	scheduled	55.14	645.9	58.8	1168	803	51.20	4.17

Confirmation + switch-point sweep (4 seeds, 7,000 ticks, 16 runs; Table 9–10, Fig. 6). The pilot result reverses: **the plain shooting baseline wins every aggregate metric across four seeds** (peak 86.7 $\pm$ 8.4 vs 72–77; final 61.5 $\pm$ 4.7 vs 44–59; most planting; fastest). Within the scheduled family earlier switches beat later ones (4k > 5k > 6k — sched@6k even collapses at seed 1: peak 50.2, 443 seeds planted), but no switch

point closes the gap. The 2-seed pilot win sits comfortably inside per-seed swings (baseline peaks range 77–100; sched@6k 50–83). Warmup gating is implemented correctly and its internal ordering is sensible — the heavy machinery it protects simply is not worth its cost on this task at this model quality.

TABLE 9. Warmup confirmation + switch-point sweep, complete per-run data (4 seeds, 7,000 ticks). sched@N = P1 until tick N, then CEM+imagination.

seed	arm	peak	lifespan	tiles	EAT	seeds	final	ticks/s
1	baseline	100.11	794.6	100.9	4154	3112	61.55	6.48
1	sched@4k	82.99	743.5	77.8	2992	2758	54.54	5.56
1	sched@5k	79.70	743.4	53.1	2754	2771	59.54	5.52
1	sched@6k	50.24	621.1	87.2	543	443	40.58	6.32
2	baseline	84.49	745.9	80.8	2930	2499	55.04	6.31
2	sched@4k	66.83	684.7	49.6	1424	1358	55.59	5.07
2	sched@5k	74.96	751.5	59.8	2234	1673	52.83	5.75
2	sched@6k	82.70	740.8	80.6	2825	2261	40.67	6.35
3	baseline	76.97	744.2	57.3	2102	2457	61.13	7.23
3	sched@4k	81.16	765.3	92.4	2385	2029	62.35	5.35
3	sched@5k	77.83	743.5	76.3	2508	2235	51.82	5.37
3	sched@6k	72.97	779.7	77.0	1934	1599	47.16	5.96
4	baseline	85.06	788.5	95.2	2265	2600	68.27	7.14
4	sched@4k	78.49	762.5	48.2	2671	1767	62.37	5.10
4	sched@5k	66.71	691.6	69.0	1146	1243	49.94	5.73
4	sched@6k	82.73	743.2	92.6	3179	3142	49.22	6.00

TABLE 10. Sweep aggregates (mean $\pm$ std over 4 seeds).

Arm	peak fitness	final fitness	seeds planted	ticks/s
baseline (shooting)	86.7 $\pm$ 8.4	61.5 $\pm$ 4.7	2667 $\pm$ 262	6.8
sched@4k	77.4 $\pm$ 6.3	58.7 $\pm$ 3.7	1978 $\pm$ 510	5.3
sched@5k	74.8 $\pm$ 5.0	53.5 $\pm$ 3.6	1981 $\pm$ 576	5.6
sched@6k	72.2 $\pm$ 13.3	44.4 $\pm$ 3.9	1861 $\pm$ 985	6.2

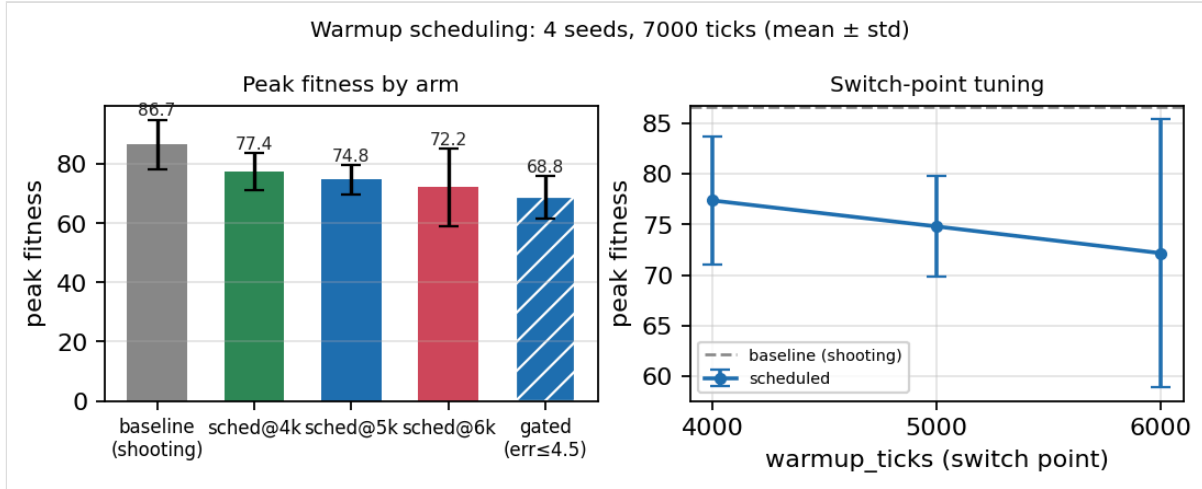


FIGURE 6. Warmup confirmation + switch-point sweep (4 seeds, 7,000 ticks, mean $\pm$ std). Left: peak fitness by arm; the hatched bar is the readiness-gated variant (switch on measured error ≤ 4.5 , Sec. 6.7). Right: the fixed-switch family against the baseline reference — earlier switches are better, none reach the baseline.

6.7 Measuring the model itself

The consistent pattern of Sec. 6.4–6.6 — the sharper the search, the worse the outcome — points at compounding model error rather than at the search algorithms. Every planner variant rolls the dynamics head open-loop for D steps, yet the head is trained

only on one-step prediction; nothing constrains the regime the planner actually consumes [18,19]. We therefore (i) measure that regime directly during training and (ii) train it directly. **Setup:** two arms, seeds 1–3, 6,000 ticks each (identical to the sweep baseline: shooting planner, world-model head on) — 1-

step, the standard auxiliary loss; *multi-step*, plus the k-step consistency loss (k=3, horizons 2–3, coefficient 0.5). Both arms

log the open-loop rollout-error diagnostic (k=3) every learner update; the population mean is recorded each generation.

TABLE 11. Open-loop rollout error (latent MSE at horizon k=3), complete per-run data: population mean at each generation boundary, 3 seeds per arm, 6,000 ticks.

tick	1-step s1	s2	s3	mean $\pm$ std	multi-step s1	s2	s3	mean $\pm$ std
1000	6.49	6.50	6.69	6.56 ± 0.09	6.24	6.02	6.00	6.09 ± 0.11
2000	6.16	6.03	5.90	6.03 ± 0.11	6.12	5.94	6.19	6.08 ± 0.11
3000	5.91	5.17	5.35	5.48 ± 0.31	5.85	5.20	5.73	5.59 ± 0.28
4000	5.23	4.88	4.25	4.79 ± 0.40	5.04	4.18	4.59	4.60 ± 0.35
5000	4.16	3.49	3.28	3.64 ± 0.37	3.81	4.20	3.71	3.91 ± 0.21
6000	2.64	3.49	2.54	2.89 ± 0.43	2.22	3.42	3.15	2.93 ± 0.52

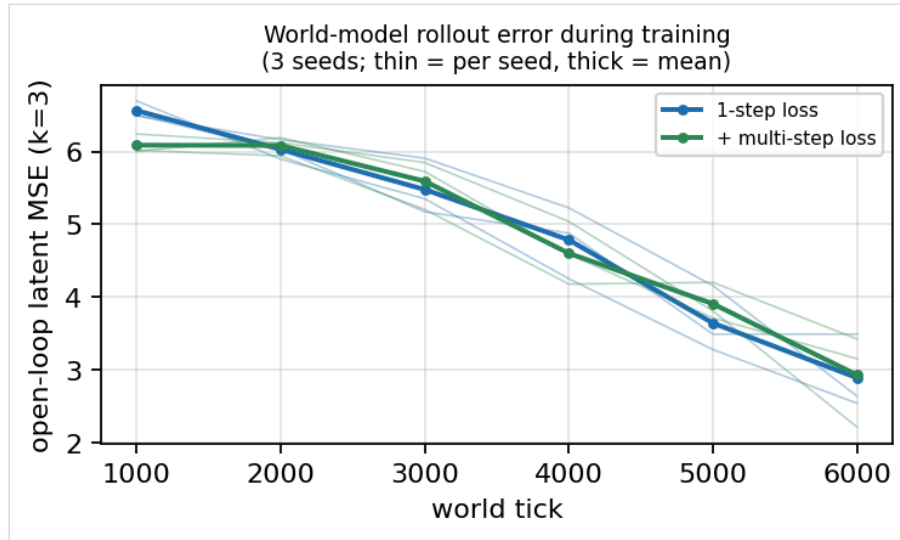


FIGURE 7. Open-loop rollout error (k=3) during training, three seeds per arm (thin lines) with per-arm means (thick). Error falls monotonically in all six runs; the multi-step loss separates only at the start — exactly where the model is worst.

Three findings. (i) The diagnostic works, and it quantifies the cold-start problem. Open-loop error falls monotonically in all six runs — $6.56 \rightarrow 2.89$ (–56%) under the standard loss. At tick 1,000 the error is **2.3 $\times$ its tick-6,000 level**: the measured version of the warmup hypothesis — early planning really does consume a much worse model. Critically, the error is *still falling* at tick 6,000; the model never “finishes” training at these horizons, consistent with no fixed switch point working in Sec. 6.6. **(ii) The multi-step loss helps exactly where the model is worst.** At tick 1,000 the multi-step arm’s error is 6.09 ± 0.11 vs 6.56 ± 0.09 — a 7% reduction, and the only measurement point where the two arms’ seed ranges do not overlap (6.00–6.24 vs 6.49–6.69). From tick 2,000 on the arms are statistically indistinguishable: multi-step training accelerates early open-loop competence — the regime a cold-start planner consumes — without changing converged model quality at this capacity. **(iii) No downstream fitness change** (the no-harm check): final fitness 63.2 ± 10.7 (1-step: $48.2/72.0/69.5$) vs 56.2 ± 6.0 (multi-step: $56.3/63.4/48.7$); peak

66.1 ± 6.7 vs 57.4 ± 7.6 — overlapping ranges at $n = 3$, no benefit and no clear harm. Together with Sec. 6.6 this says early model error is real and reducible, but *not the only* binding constraint on model-based planning here.

Downstream test of the readiness gate. Finally, we A/B the gate itself: the scheduled planner re-run at the sweep’s exact conditions (seeds 1–4, 7,000 ticks) but switching on the **measured** per-agent error EMA (threshold 4.5, picked where the mean curve of Fig. 7 crosses $\approx$ the sweep’s best fixed point; deadline 6,000). Table 11b gives every run. Aggregate: peak **68.8 ± 7.4** , final 49.9 ± 8.5 — **0/4 seeds beat the baseline** ($86.7 \pm 8.4 / 61.5 \pm 4.7$), and the mean trails even the fixed sched@4k (77.4 ± 6.3 ; the hatched bar in Fig. 6). The gate behaved exactly as designed, and its internal pattern re-confirms the sweep’s ordering: the seed whose error crossed earliest (seed 2, $\approx 2,500$) produced the best gated run (81.5), while the two seeds that never crossed rode the 6k deadline — the worst fixed switch point. Two structural effects kept switches late: at threshold 4.5 the population-mean error stays

high for most of the run, and the gate is *per-agent* — newborns start with a fresh error EMA, so population turnover continually re-enters warmup (visible as the gated arm's higher ticks/s: it pays for CEM + imagination less often). Raising the threshold can only converge toward sched@4k, which still loses. Across

three independent designs — the 20-run replication, the 16-run sweep, and this 4-run gated A/B — **no gating policy, fixed or measured, makes CEM + imagination competitive with plain shooting on this task.**

TABLE 11B. Readiness-gated scheduled planner, complete per-run data (switch when per-agent rollout-error EMA ≤ 4.5 , deadline 6,000; seeds 1–4, 7,000 ticks).

seed	peak	lifespan	tiles	EAT	seeds	final	ticks/s	pop-mean err crosses 4.5
1	64.43	720.1	98.9	2045	1358	37.27	6.81	$\approx$ tick 4,700
2	81.49	773.9	98.2	2420	1966	60.92	7.22	$\approx$ tick 2,500 (earliest)
3	65.65	715.4	65.8	1888	1718	48.91	6.92	never — 6k deadline
4	63.66	708.1	44.4	1211	909	52.32	8.08	never — 6k deadline

6.8 Population dynamics across founder counts

Three v3.5+PPO worlds (48×48, cap 60) seeded with 8, 24, and 48 founders (distinct seeds). Table 12 samples the trajectories; Fig. 8 shows the full curves. All three converge to the same

carrying capacity (the cap) within 100–200 ticks regardless of founder count while fitness keeps rising: the ecology self-regulates, and population size is set by the environment, not the initial condition — the stability prerequisite for the long-horizon studies above.

TABLE 12. Population and mean fitness vs. tick for three founder counts (sampled; full curves in Fig. 8).

tick	pop (8 f.)	fitness	pop (24 f.)	fitness	pop (48 f.)	fitness
0	8	0.0	24	0.0	48	0.0
100	32	4.05	60	5.83	60	6.99
200	60	4.70	60	9.11	60	11.58
400	60	7.84	60	9.55	60	8.83
600	60	10.35	59	12.84	60	11.33

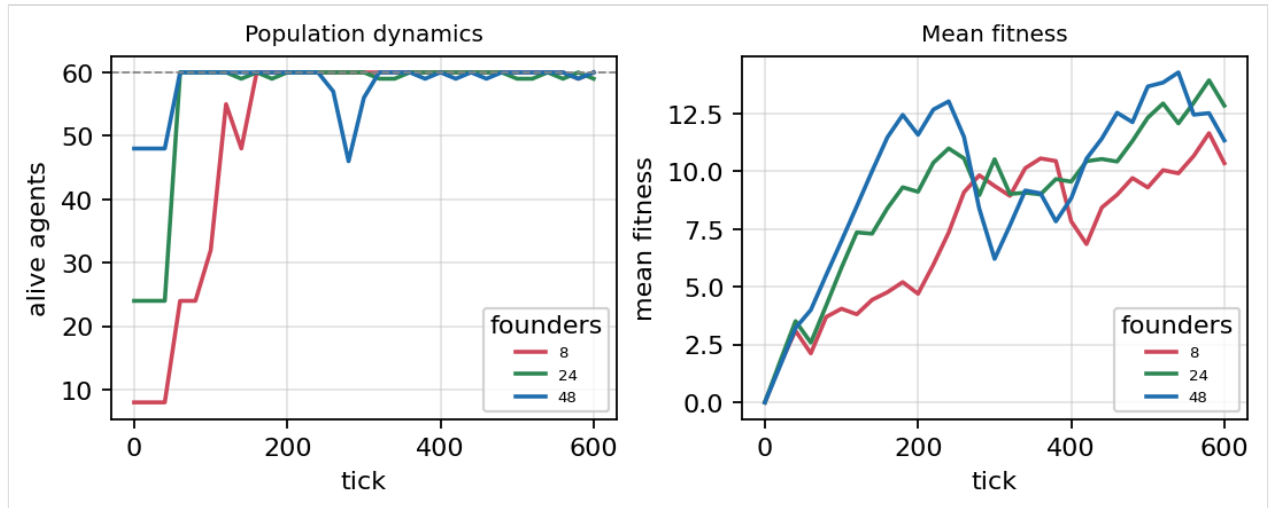


FIGURE 8. Population (left) and mean fitness (right) vs. tick for three founder counts. Different starts, same carrying capacity.

7. Discussion

A weak model is useful for cheap things and harmful for expensive ones. The same 20%-better-than-nothing model family produced a genuine behavioural phase change when used for blunt search and surprise signals (Sec. 6.3), a small

robust gain when used to keep rollouts in-distribution (P1, Sec. 6.5), and consistent losses when used as the substrate for concentrated optimisation (CEM, imagination; Sec. 6.5–6.6). Ranking a handful of action candidates needs only the model's gradient of "better vs worse"; CEM's elites and imagination's training targets need its *absolute* predictions several steps out

— precisely where one-step training gives no guarantees. Sharper search amplifies model error instead of value [19].

Scheduling cannot rescue a model that stays weak. The warmup results order correctly (later switches are worse — less time to amortise the expensive controller) but never cross the baseline: at no switch point does the model become good enough for CEM+imagination to pay their bills. The structural reasons are visible in the sandbox: 9 discrete actions and depth-2 horizons mean uniform shooting already covers the search space, and fitness is dominated by reactive behaviours a greedy policy discovers anyway.

Measure, then trust. The toolkit of Sec. 5.6 turns "the model is probably bad early" from a hypothesis into a logged per-generation quantity, gates the expensive machinery on that quantity rather than a guess, and trains the model in the regime the planner consumes. The measurement (Sec. 6.7) both vindicates and reframes the warmup idea: at tick 1,000 the model's rollout error is $2.3\times$ its tick-6,000 level — the cold-start problem is real — but the error is still falling when every fixed switch point of Sec. 6.6 fires, so the gate was pointing at the right quantity and that quantity never reached "good." The gated A/B closes the loop: switching on the measured error behaves sensibly (earliest crossing $\rightarrow$ best gated run) yet still loses to never switching at all — the right gate on machinery that does not pay for itself is still a loss. This is the infrastructure the earlier experiments should have had, and it is how we can now say the negative result is about the machinery, not the schedule.

Replication is not optional. Two effect sizes of +25–32% (P2, P3) and one 2-of-2-seed win (scheduled) all evaporated under 4 seeds. Ecological simulations with reproduction, death, and weather have enormous seed variance — baseline peaks

alone ranged 77–100 across seeds in the sweep. Any single-seed claim in such systems should be read as an anecdote.

8. Limitations

$n = 4$ seeds is still small: the sweep's baseline-vs-scheduled gap is $\sim 1.4\sigma$ — robust in *direction* (baseline $\geq$ every scheduled arm on every metric) but not a formal significance result. Horizons are short (2k–7k ticks), worlds small (48–160 per side, ≤ 100 agents), and fitness is an in-silico proxy. The dynamics head is a deterministic MLP; the planner ignores predictive uncertainty (no ensemble). Sec. 6.3 toggles planner and curiosity together. The multi-step-loss experiment measures model error, not downstream fitness, and its $n = 3$; the readiness-gate A/B is $n = 4$ at a single, untuned threshold (4.5).

9. Conclusion

We built a reproducible multi-agent evolutionary sandbox, measured what a small learned world model can and cannot do in it, and reported the full arc including the reversals: architecture selection by competition works; a weak model powers a real behavioural phase change; policy-guided rollouts survive replication; CEM, imagination, and warmup scheduling do not; and the model's open-loop rollout error — now measured on-line and trainable directly — is the quantity that decides which of these outcomes you get. Reproducibility: Sec. 6.1 `scripts/competition_run.py --ticks 4000`; Sec. 6.2 `main.py --world-model-log` then `scripts/train_world_model.py`; Sec. 6.3–6.6 matched `main.py --no-viz --learning --mode rl` runs over the `config/planning*_v35.yaml` configs at the stated seeds/lengths; Sec. 6.7 `config/wm_quality_{1step,multistep}_v35.yaml` with `--metrics-csv` (column `wm_rollout_error`), and the gated A/B `config/planning_sched_gated_v35.yaml`; analysis `scripts/analyze_logs.py`; figures `paper/make_figures.py`. All per-run data appears inline in Tables 6, 8, 9, 11, and 11b.

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This report describes only implemented, measured functionality of the Emergent World-Model Sandbox. Every per-run number is inline in the tables above; figures are generated from real simulation state and committed run artifacts (`paper/make_figures.py`). Cited works are the sources of ideas adapted here; the implementations are deliberately simplified and any shortfalls are the author's.